

Gadiel Moisés Fonseca Ramírez

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EDUCATION

National Autonomous University of Mexico (UNAM)

Mexico City

Bachelor of Computer Engineering

Relevant Coursework: Data Structures & Algorithms, Statistical Analysis & Data Mining, Combinational & Sequential Logic Circuits Design, Microcontroller & Embedded Systems Programming with Multi-Addressing Modes, Computer Graphics & 3D Modeling, Computer Networks & Cybersecurity, Operating Systems, Database Systems.

SKILLS

Java (21/17/8), Groovy, C, C++, Python, SQL, SOQL, JavaScript, TypeScript, Bash, HTML (Markup), CSS (Styling), XML; Agilent SLIMS, Sapio LIMS (Biobanking, Freezer Sample Management, Plugin Architecture, System Rules); Spring Boot, Spring Security, Spring Data JPA, Hibernate, React, React Router, Vite, Bootstrap, Tailwind CSS, Lombok, Mendix (Low-Code Platform); Verilog, VHDL, FPGA Design, Microcontrollers, Analog-to-Digital Converters (ADC), UART, I2C, SPI; PostgreSQL, Oracle DB, HeidiSQL, SQL Developer, Flyway (Schema Migrations), Relational Schema Design, JSON, Amazon S3; Linux (Debian-based, Arch-based), Docker, Docker Compose, Cloudflare Tunnels, Apache HTTPD, Apache Tomcat, systemctl, SSH, SMTP (Email Services), AWS (S3, SNS, SQS), Firebase Authentication (JWT, Google OAuth), Apache Maven, Gradle, Git, Bitbucket, GitHub, CI/CD Pipelines, pnpm Workspaces; Splunk, Postman, FFmpeg, OpenAPI/Swagger, IntelliJ IDEA, Visual Studio Code, Eclipse; Agile/Scrum, Sprint Planning, Jira, Jira Tempo, Code Reviews, Six Sigma; Spanish (Native), English (C1 Advanced — EF SET Score 70/100), French (A2 Elementary).

CERTIFICATIONS

- **EF SET Standard English Certificate:** 70/100 (C1 Advanced) — Verified Professional Fluency (2024)
- **Alura Latam Professional Certificates:** Git & GitHub (2024) · HTML5 & CSS3 (2024) · Programming Logic with JavaScript (2024)
- **Google Cloud Computing Foundations:** Cloud architecture, infrastructure services, and storage solutions (2023)
- **Six Sigma Green Belt:** Quality management, statistical process optimization, and defect reduction (2021)
- **Macquarie University / Coursera:** Excel Skills for Business: Essentials (2021)

PROJECTS

Planeta Gema — Full-Stack Gemstone & Jewelry E-Commerce Platform — *Personal Project*

<https://planetagama.com> | *Repository:* [GitHub \(Private\)](#)

- Architected and deployed a production full-stack multi-vendor e-commerce platform (Java 21 / Spring Boot 3, React 19 / Vite 6, PostgreSQL 16, Docker Compose) with Firebase JWT RBAC, merchant data isolation, and automated daily financial settlement reports.
- Engineered a JSON-backed multi-tier pricing engine with JPA compounding lifecycle hooks, an adjacency-list hierarchical catalog tree reducing backend code by 60%, and a server-side FFmpeg pipeline cutting image payloads by 85%.

PROFESSIONAL EXPERIENCE

Zifo RnD Solutions

Mexico City (Client: Regeneron Pharmaceuticals)

Scientific Application Analyst & Software Engineer

May 2025 – September 2026

- Developed and deployed modular Java plugins (fat JARs via Gradle/Maven) and Groovy rules for Agilent SLIMS and Sapio LIMS, automating storage and fulfillment workflows for over 6,000,000 biological sample tubes in automated freezers.
- Architected enterprise migration safeguards during cutover to Sapio: engineered Groovy system rules, API 403 Forbidden interceptors, and JSON payload validation to enforce strict read-only states across 6M+ records with zero data corruption.
- Standardized inter-departmental REST API JSON schemas with Postman test suites; refactored heavy scientific query plugins into asynchronous worker threads, resolving browser timeout warnings and duplicate technician submissions.
- Administered multi-tier Linux application instances (Apache HTTPD, Tomcat, Splunk) and established Zifo's centralized global LIMS starter codebase, accelerating engineer onboarding while leading daily Scrum standups across three global time zones.

Siemens DI SW

Mexico City

Software Engineer Intern

May 2024 – April 2025

- Developed a modular REST API using Java 17 and Spring Boot, implementing Spring Data JPA with DTO models for secure multi-database connectivity and Maven multi-profile builds for staging/production deployment.
- Engineered an interactive front-end data visualization platform in Mendix, consolidating disparate Oracle DB and Salesforce datasets with dynamic filtering, sorting, custom query views, and exportable reports.
- Automated incoming data request processing by configuring SMTP email services and scheduled background tasks to monitor inbound email streams, trigger transactional database updates, and dispatch automated alerts.
- Maintained disciplined version control and repository hygiene using Git and Bitbucket, facilitating seamless pull request reviews, automated build validations, and zero-regression production deployments.